Operating Systems

The structure of a Computer system can be visualized as having four basic components:
1. Hardware – Provides basic computing resources - CPU, memory, I/O devices
2. Operating system – Controls and coordinates use of hardware among various applications and users
3. Application programs – Define the ways in which the system resources are used to solve the computing problems of the users
4. Users – Anybody who desires work to be done by a computer system. People, machines, other computers



Types of OS

OS are classified into following different types depending on their capability of processing.

Single User and Single Tasking OS:

These are simple operating system designed to manage one task at a time ,for use by a single user for a standalone single computer for performing a single task .

Single User and Multitasking OS:

These OS allow execution of more than one task or process concurrently by dividing the processor time amongst different tasks.

Multi-programming OS:

These OS allow more than one programs to run at the same time .

Real Time OS:

These are designed to respond to an event within a predetermined time. These operating systems are used to control processes

Embedded OS:

Embedded in a device in the ROM. They are specific to a device and are less resource intensive.

Function Of OS

The OS performs basic tasks such as controlling and allocating memory, prioritizing system requests, controlling input and output devices, facilitating networking, and managing files

Introduction to UNIX OS

- Unix is an OS for Programmers as shell(the command interpreter)provides the programming facility.
- It provides an in-built security mechanism through the user name and password, combined with the access rights associated with files
- Developed by Ken Thompson and Ritchie originally in assembly, and later in C, thus making it portable to other machines

Supports C, Fortran, Basic, Pascal, COBOL, Lisp, Prolog, Java, Ada compilers

Features of Unix Operating System

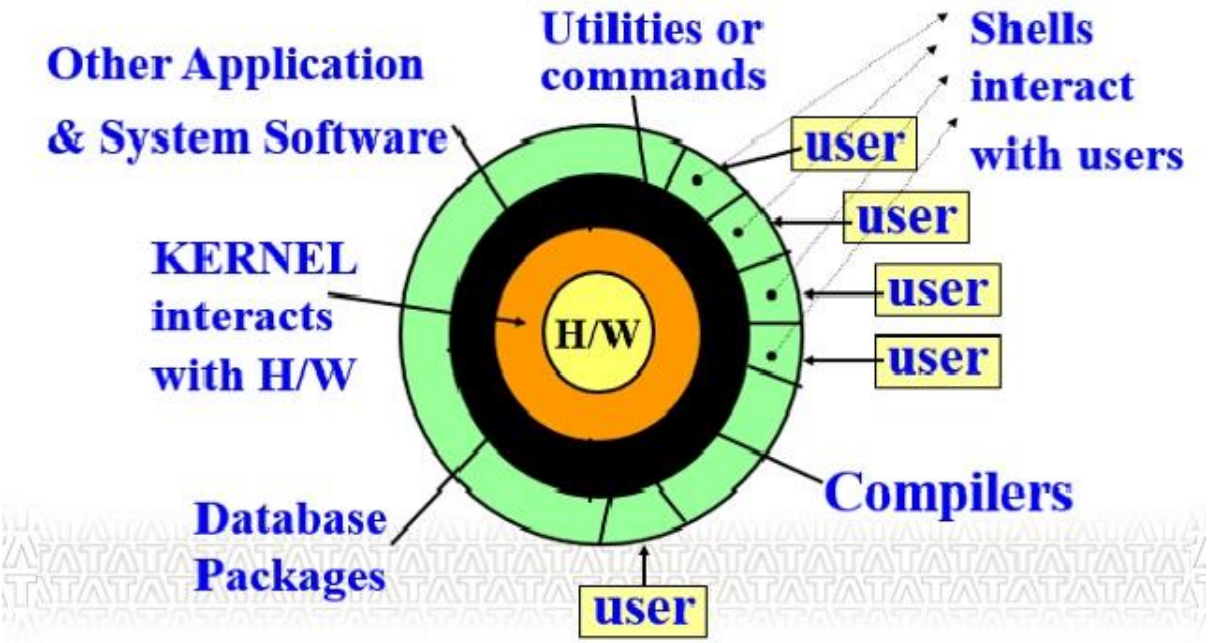
• Multi-user and Multitasking
• Everything is a file
• Configuration data stored in text
• Small, single purpose programs
• Ability to chain programs together to perform complex task
• Facility of background processing

Architecture of the UNIX System

The UNIX operating system has a layered architecture having three main components
• Kernel

• Shell

• Utilities



The Unix Kernel

Kernel is a collection of programs mostly written in C which allocate the system resources and coordinate all the details of the computer's internals.

Functions of Kernel:

- It allocates time and memory to programs and handles the file store and communications
- Interacts directly with the hardware through device drivers
- Provides sets of services to programs
- Manages memory, controls access, maintains file system, handles interrupts, allocates resources of the computer

System calls

The system calls are functions used in the kernel itself. UNIX system calls are used to manage the file system, control processes, and to provide interpocess communication.

System calls can be categorized as:

- File structure related calls -For example create, open,read, write,lseek,dup etc.
- Process Related calls -For example fork,exec,wait,exit etc
- Inter process related calls - For example pipe,msgget,msgsnd etc.